Mohamed Hedi Chebil

Data, BI & AI Engineering Student — 6-month end-of-study internship from January 2027

Tunis, Tunisia | chebilmedhedi0@gmail.com | +216 93 964 853 | linkedin.com/in/mohamed-hedi-chebil-817a49404 | github.com/hedi216214 | hedi216214.github.io

PROFILE

Final-year engineering student (ESPRIT, Business Intelligence & Big Data). Industry experience in star-schema data warehousing, Talend/SSIS ETL and Power BI (advanced DAX); LLM-based Text-to-SQL assistant with RAG (Chroma), a read-only guard and 23 tests; deep learning road-obstacle detector (PyTorch, U-Net, ResNet) deployed in the browser with ONNX. Availability: 6-month end-of-studies internship from January 2027 (February possible).

PROFESSIONAL EXPERIENCE

Data & BI Intern — AXIA SOLUTION, Tunis — Jun – Aug 2026

- Designed a target Data Warehouse architecture in star schema (fact and dimension tables) centralizing multi-source data.
- Built complex ETL workflows with Talend Open Studio and SSIS: cleaning, transforming and consolidating heterogeneous sources.
- Power BI dashboards for management: data model optimization, advanced DAX; translated business needs into KPIs.

Engineering Intern — LOUPYTECH, Tunis — Jul – Aug 2025

- Client monitoring portal: user management and Centreon supervision reports.
- Spring Boot back-end and REST API, database schema, secure authentication; Angular front-end; API testing with Postman.

Discovery Intern — Big Data & BI Department, AXIA SOLUTION — Summer 2024

- Data governance, organization of storage and analytics platforms.

PROJECTS

Text-to-SQL BI Assistant (LLM, RAG) — github.com/hedi216214/text-to-sql-bi-assistant

- Business questions in FR/EN translated to SQL over a retail star-schema DW (SQLite, 60,000 sales lines); RAG: 32 question → SQL examples in a Chroma vector database, the 3 most similar added to the prompt; one automatic correction attempt on SQL errors.
- Read-only guard (single SELECT, read-only connection, SQLite authorizer limited to the 5 tables, 5 s timeout, 200-row cap); OpenAI-compatible APIs (Groq, Mistral, OpenAI, local Ollama); evaluation on 20 questions with reference SQL (execution accuracy) + 23 offline pytest tests.

Vigie Chaussée — Road Obstacle Detection (Deep Learning, PyTorch) — github.com/hedi216214/vigie-chaussee

- ESPRIT team project (6 students, CRISP-DM), RoadAnomaly dataset: 9 models on 3 objectives (detect, localize, passable lane); transfer learning won each time: ResNet-50 F1 0.82, U-Net ResNet-34 IoU 0.70 (0.41 from scratch).
- My role, deployment: 3-model chain, one ONNX graph with int8 quantization (143 → 36 MB, IoU kept at 0.70); serverless web app (onnxruntime-web) analysing a photo in 2–3 s on CPU.

Retail Sales Optimization — ESPRIT academic project (parapharmacy)

- Talend ETL pipeline into a PostgreSQL Data Warehouse (2+ years of sales); Power BI KPIs on stock, margin and B2B customers.
- Python forecasting, segmentation and anomaly detection models (85% forecast accuracy on test data), served via FastAPI; MLflow, Docker; Angular front-end via Flask API.

SKILLS

BI & Data Warehousing: Power BI, DAX, Power Query, Talend, SSIS, dimensional modeling, star/snowflake schema, Data Warehouse, OLAP cubes
AI / ML: Python, PyTorch, TensorFlow, Keras, CNN, U-Net, ResNet, transfer learning, image segmentation, LLM, Prompt Engineering, Text-to-SQL, RAG, ChromaDB, forecasting, classification, anomaly detection
Data Engineering & MLOps: Apache Spark, Spark SQL, MLflow, FastAPI, Docker, ONNX, Git, pytest
Databases: Advanced SQL, PostgreSQL, MySQL, Oracle SQL, SQLite
Applications & methods: Microsoft Dynamics 365, Power Platform, Spring Boot, Angular, REST API, PHP Symfony; Agile, Scrum, MVC, CRISP-DM

EDUCATION

Engineering degree in Computer Science, BI & Big Data specialization — ESPRIT, Tunis — 2024 – 2027 (expected)

- 2026–2027 courses: Spark SQL, Spark Streaming, Elasticsearch, Deep Learning with TensorFlow (CNN, RNN, Autoencoders), NLP, Transformers, LLM, GNN, MLOps, DevOps, Microsoft Dynamics 365 CRM, SAP BW/4HANA.

Integrated preparatory cycle — École Polytechnique de Sousse — 2022 – 2024
Technical Baccalaureate — Lycée Othman Chatti, Msaken — 2018 – 2022

CERTIFICATIONS

Hedera Certified Developer Associate (HCDA), 2026 · Blockchain Certification, 2026 · AWS Academy Cloud Foundations (in progress)

LANGUAGES

French (C1), English (C1), Arabic (native)